COURSE CODE: DECAP780

COURSE NAME:PROBABILITY AND STATISTICS

Short Questions(2 Marks)

Q1. Differentiate between samples and populations in the context of statistics.
Q2.Define mathematical expectations and their significance in probability.
Q3. What are moments of random variables, and how are they calculated?
Q4.Describe two-way classification in analysis of variance.
Q5. Explain the Poisson distribution and its characteristics.
Q6.What is statistical quality control and its role in process control?
Q7. Define time series in statistics.
Q8. How is hypothesis testing used in business contexts?
Q9. What is the F-statistic, and how is it critical in the F-distribution?
Q10. Name some statistical software tools used in analysis.
Q11. What is Statistical Inference, and what are its types?
Q12. Explain statistical interference with the help of example.
Q13. Explain the absolute measure of dispersion with the help of example.
Q14. Define correlation and explain its significance in statistical analysis.
Q15. Explain the concept of regression.
Q16. Briefly explain the properties of moment generating function.
Q17. Give an example of when you would use a c chart in process control.
Q18. Differentiate between sampling with and without replacement.
Q19. Describe the importance of checking for homogeneity of variances in ANOVA. What test is commonly used for this purpose?
Q20. What is the fundamental principle of Bayesian statistics?

Q21. What is permutation?
Q22. Define combination.
Q23. What is probability distribution?
Q24. Define binomial distribution.
Q25. What is normal distribution?
Q26. Define hypothesis testing.
Q27. What is null hypothesis?
Q28. Define alternative hypothesis.
Q29. What is sampling?
Q30. Define population in statistics.

Long Questions(10 Marks)

Q1. Execute Various measure of descriptive statistics used by researchers along with suitable example of each?
Q2. Execute various measures of variability and dispersion used by researchers along with example?
Q3: Explain the concept of raw moments & central moments, along with suitable example?
Q4. Explain the concept of Effect of change of origin and scale on moments with suitable example?
Q5. Discuss the importance of index numbers in measuring changes in economic variables over time.
Q6. Compare and contrast the different methods of constructing index numbers, such as the Laspeyres, Paasche, and Fisher indices. Highlight their advantages and disadvantages.
Q7. Define skewness and Kurtosis for continuous probability distribution.
Q8. Define Variance, Properties of variance and covariance?
Q9. Define what is normal distribution and example?
Q10. Define what is difference between Binomial, Poisson and Negative Binomial Distribution
Q11. Define a time series and its importance in data analysis. Give examples of economic and financial time series.
Q12. Break down the components of a time series, including trend, seasonality, and irregular variations. Explain how these components can be identified and analyzed.
Q13: Explain the concept of conditional probability along with suitable example?
Q14: Explain the concept of conditional probability along with suitable example?
Q15. Discuss use of Index numbers and process to create index numbers?
Q16. Illustrate use of time series data and what are the different type of time series analysis?
Q17. Explain the difference between random sampling with replacement and random sampling without replacement. Provide real-life scenarios where each method would be appropriate
Q18. Define a hypothesis and its role in scientific research. Describe the steps involved in hypothesis testing, from setting up hypotheses to making a decision.
Q19. Explain moments of two or more random variables and their practical significance.
Q20.Explain the objectives of analyzing variance in different scenarios.
Q21.Elaborate on Bayes' theorem and provide a real-world scenario where it can be applied effectively.
Q22.Discuss the properties of expectations and their role in probability theory.
Q23. Explain the concepts of random samples and random numbers in sampling theory.
Q24. Explain the Student's t-distribution, its properties, and critical values.
Q25.Explain the role of the Data Analysis Tool Pack in Microsoft Excel. Provide examples of situations where it can be particularly useful for data analysis tasks.
Q26. Discuss the characteristics and properties of the binomial, Poisson, negative binomial, and normal distributions,
Q27.Describe the advantages of using R-Studio for statistical analysis, including any unique features or packages it offers.
Q28.Discuss Bayesian concepts and inferences in statistical analysis.
Q29 Explain Pearson's correlation coefficient and discuss its properties. How is it calculated, and what does it indicate about the strength and direction of the relationship between variables?
Q30. Describe How Bayes Estimates are Derived Using the Decision Theory Framework, Including the Role of Loss Functions.